

ZOMATO RESTAURANT DATA ANALYSIS REPORT

1. Project Title

Zomato Restaurant Data Analysis Using SQL and Python (EDA Project)

2. Objective

The objective of this project is to analyze Zomato restaurant data to extract meaningful insights regarding:

- Restaurant distribution across cities
- Customer rating patterns
- Cost trends of restaurants
- Popular cuisines
- Relationship between cost, rating, and votes

The goal is to understand customer preferences and restaurant market trends using data analysis techniques.

3. Dataset Description

The dataset used in this project contains information about restaurants listed on Zomato.

Key Features:

- Restaurant Name
- City
- Address
- Cuisines
- Rating (Aggregate rating)
- Votes
- Approximate Cost for Two People
- Online Order Availability
- Table Booking Option

4. Data Cleaning Process

To ensure accurate analysis, the dataset was cleaned using the following steps:

- Removed duplicate records
 - Handled missing values in important columns (Rating, City, Cuisines)
 - Standardized column names for consistency
 - Converted rating values into numeric format
 - Removed invalid or null entries
 - Ensured correct data types for numerical and categorical columns
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5. Tools & Technologies Used

- **Python** (Pandas, NumPy, Matplotlib, Seaborn)
 - **SQL (PostgreSQL)** for data querying
 - **Jupyter Notebook** for EDA
 - **Excel**
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6. Exploratory Data Analysis (EDA)

6.1 City-wise Restaurant Distribution

- Major metropolitan cities such as Bangalore, Delhi, and Mumbai have the highest number of restaurants.
 - Smaller cities show significantly lower restaurant density.
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6.2 Rating Analysis

- Most restaurants fall in the rating range of **3.0 to 4.2**
 - Very few restaurants achieve ratings above 4.5
 - Indicates moderate customer satisfaction overall
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6.3 Cuisine Analysis

- **North Indian, Chinese, and Fast Food** are the most popular cuisines
- Multi-cuisine restaurants dominate urban markets

6.4 Cost Analysis

- Majority of restaurants fall in the **budget to mid-range category (₹300–₹800 for two people)**
 - Premium restaurants are fewer in number
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6.5 Relationship Between Cost and Rating

- Weak correlation observed between cost and rating
 - Higher cost does not guarantee higher ratings
 - Customer satisfaction depends on food quality and service, not price alone
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6.6 Votes Analysis

- Restaurants with higher ratings generally receive more votes
 - Votes reflect popularity but not always quality
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7. Key Insights

- Metro cities dominate restaurant availability
 - Affordable restaurants are most common
 - North Indian and Chinese cuisines dominate the market
 - Ratings are moderately distributed, with few exceptional restaurants
 - Cost and rating have weak correlation
 - Online ordering increases restaurant reach and visibility
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8. Visualizations Used

The following charts were used for analysis:

- Bar chart → City-wise restaurant count
- Histogram → Rating distribution
- Pie chart → Cuisine distribution
- Scatter plot → Cost vs Rating
- Bar chart → Top cuisines

(All charts are included in the [/images](#) folder of the repository)

9. Project Structure

Zomato-Restaurant-Analysis/

```
|  
├── data/           # Raw dataset  
├── notebooks/      # Python EDA notebook and SQL queries file  
├── images/         # Visualizations  
├── reports/        # Final PDF report  
└── README.md
```

10. Conclusion

This project successfully analyzes restaurant data to uncover important business insights. The findings show that:

- Restaurant distribution is heavily concentrated in metro cities
- Pricing has little impact on ratings
- Certain cuisines dominate customer preferences
- Data-driven insights can help improve restaurant strategy and customer experience

This analysis demonstrates the use of **SQL + Python EDA techniques** for real-world business problem solving.

11. Author

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